**1** A student is investigating the motion of a wooden block on an inclined plane, as shown in Fig. 1.1. A falling body causes the wooden block to accelerate.

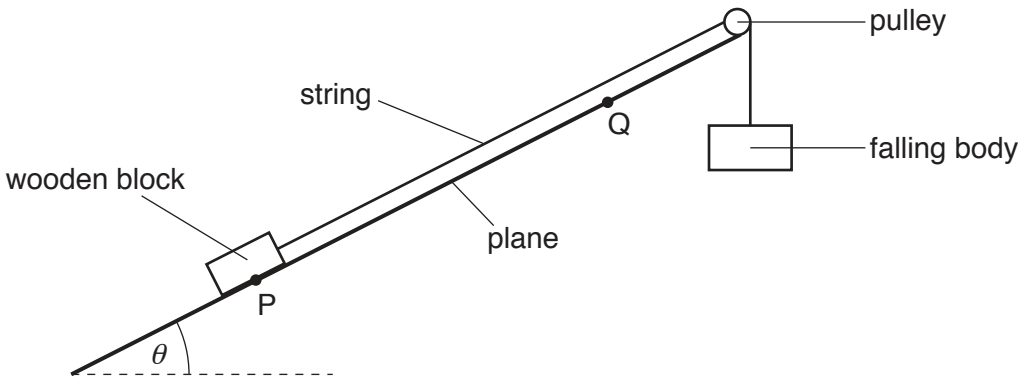



**Fig. 1.1**

The wooden block is initially at rest at point P and has velocity $v$ at point Q.

It is suggested that the relationship between $v$ and the angle $\theta$ of the plane to the horizontal is

$$\frac{(B + m)v^2}{2s} = Bg - mg\sin\theta$$

where $B$ is the mass of the falling body, $m$ is the mass of the wooden block, $s$ is the distance between P and Q and $g$ is the acceleration of free fall.

Design a laboratory experiment to test the relationship between $v$ and $\theta$. Explain how your results could be used to determine a value for $g$. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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